Part 1

Introduction to AI

Vibe Coding



Ray Villalobos
linkedin.com/in/planetoftheweb
@planetoftheweb

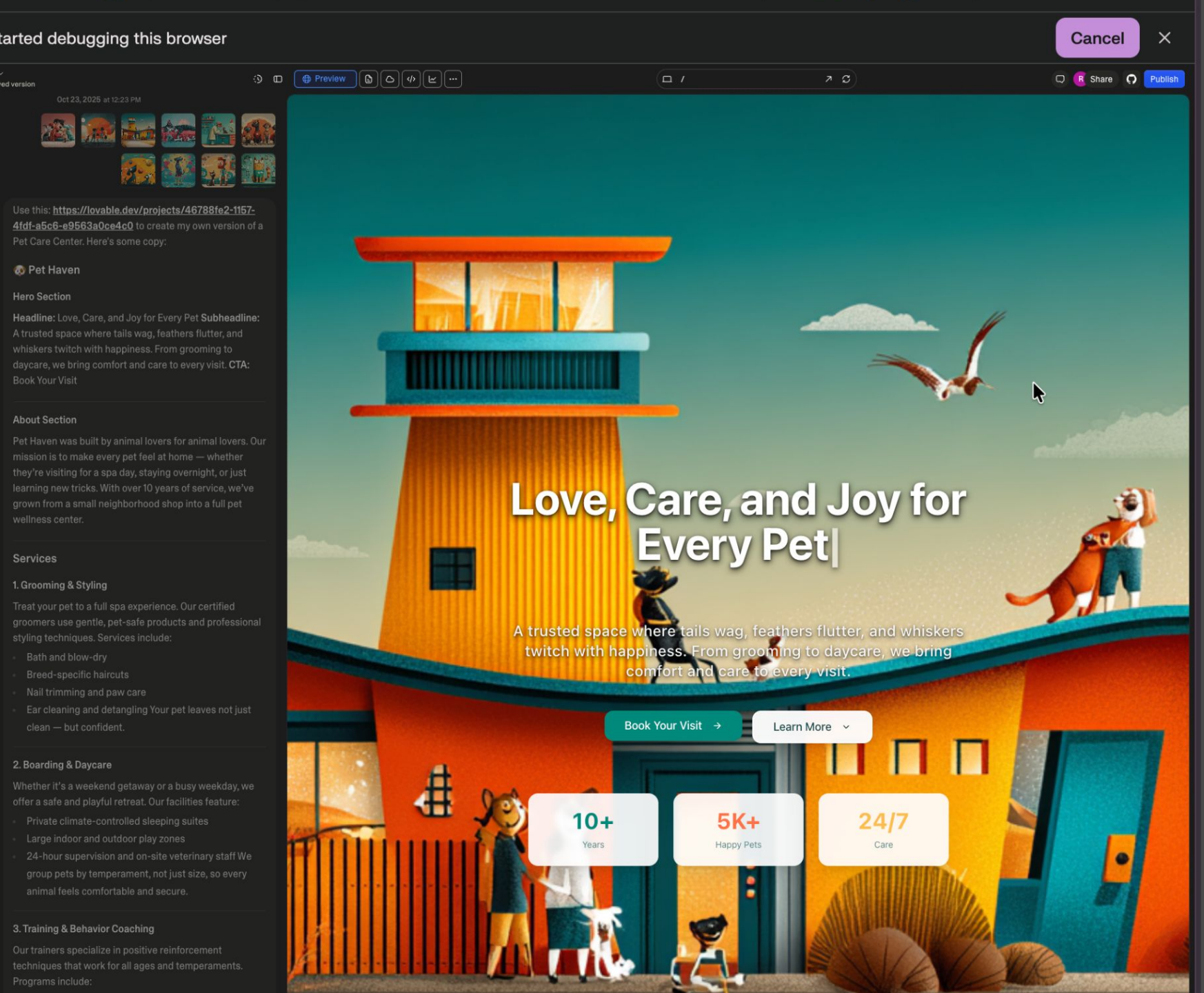




What is Vibe Coding?

February 3, 2025: Andrej Karpathy on X, coined term (OpenAI founding member, Tesla Autopilot Lead). Post hit 6.8 million views.
Merriam-Webster added the term within months.

- AI Models are now good enough that you can build apps by describing changes instead of writing code directly.
- You stop caring about the underlying code details: you accept changes without reading the code, paste errors back into the model, and let it iteratively fix things.
- The resulting codebase grows beyond your comprehension, but you can still ship working projects by treating it as “see stuff, say stuff, run stuff, copy paste stuff.”
- A new style of programming that trades traditional control and understanding for speed and friendly interaction.



What does it Mean?

You describe what you want in plain language. AI builds it with simple tools that could include chatbots, or sites like this built with Lovable.

- The skill is in the asking, not the typing.
- You are steering, not writing syntax.
- This is not about becoming a programmer. It's about becoming someone who can direct an AI to build working tools.
- You can build professional websites, tooling, data processing workflows, analysis, infographics.
- You get better at explaining what's necessary to build effective sites that do what you need them to do.
- You learn a lot about necessary programming concepts like: hosting, authentication, debugging, registration, databases, e-commerce.



Vibe Coding Spectrum

Every AI coding tool fits somewhere on a spectrum. Starting with chatbots, the no code platforms and ending with professional agentic coding tools.

- **Beginner: Chatbots.** Google Gemini, ChatGPT, Claude. Type a prompt, get a single-page output like a dashboard or infographic. No setup, No user accounts, no databases, no custom URLs, no multi-page apps, no e-commerce. Output in the chatbot until you download it.

- **Intermediate: No-Code Platforms.** Lovable, Replit Agent, V0. Full applications, not just single pages. Databases (Supabase/PostgreSQL), user authentication, Stripe payments, custom domains, multi-page routing, live deployment. Code syncs to GitHub so a developer can take over.

- **Pro Level: Developer Platforms.** Cursor, VS Code, Claude Code, Codex. AI writes code inside a professional editor. You see every file, approve every change. You choose the frameworks, hosting, database, deployment. No guardrails, no vendor lock-in. Requires comfort with a code editor.



AI Output is Non-Deterministic

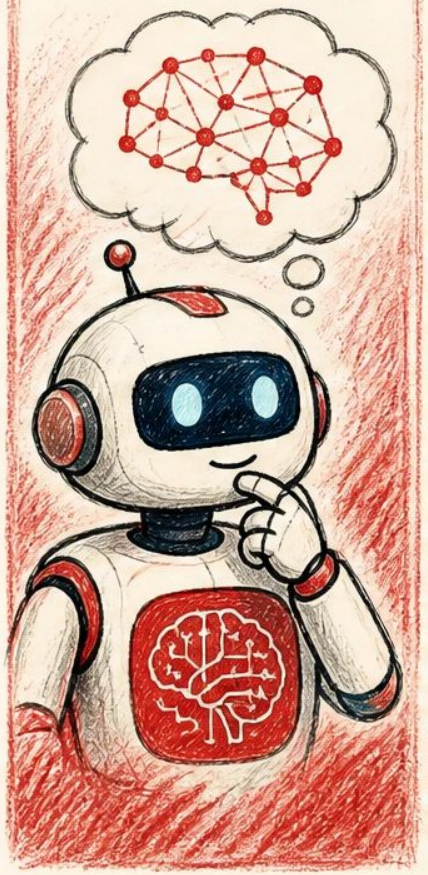
The same prompt will produce different results every time. This is a feature, not a bug, but you need to plan for it.

- AI models are probabilistic. Two people typing the same prompt will get different code, layouts, and design choices.
- Your live demo will not look identical to your rehearsal. Prepare for that.
- If you don't like what you get, regenerate or refine with a follow-up prompt.
- You can steer the output with reference images: upload a screenshot, a color swatch, or a style mockup and say "match this look."
- The more specific your prompt, the more consistent the output across runs.

1

MODEL

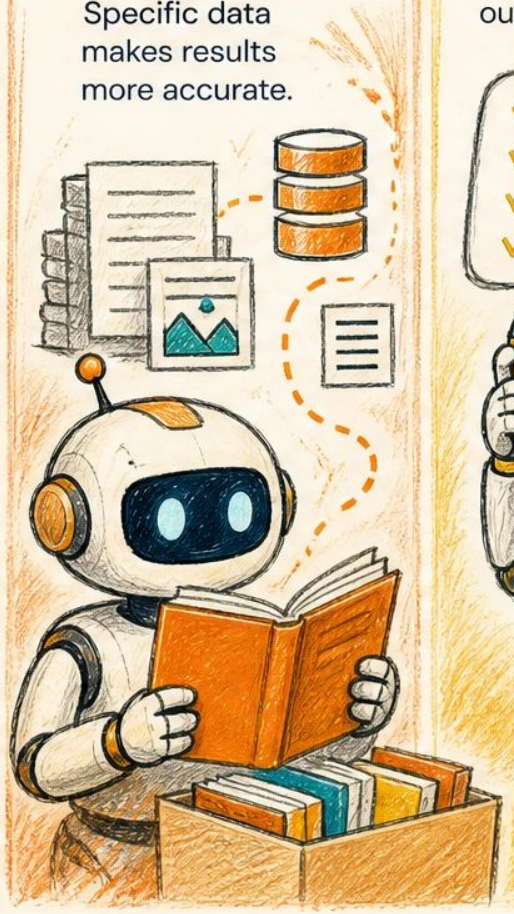
The reasoning and generation brain that makes the decisions and establishes how your prompts are answered.



2

DATA

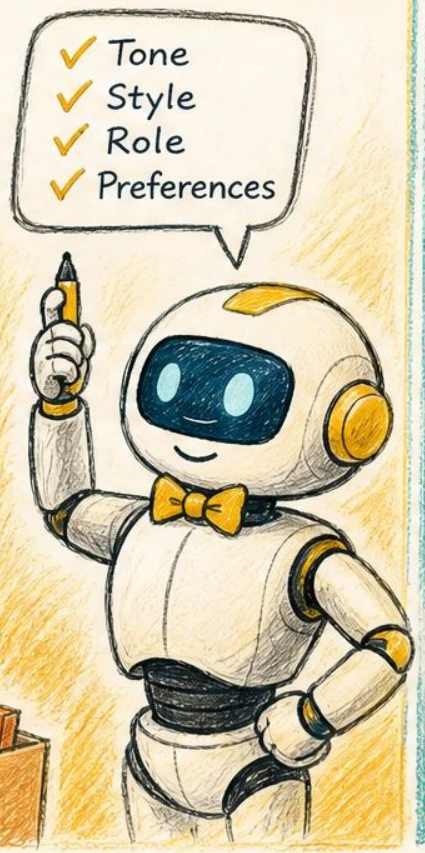
A model uses stored knowledge from training, but it's limited and time constrained. Specific data makes results more accurate.



3

PERSONALIZATION

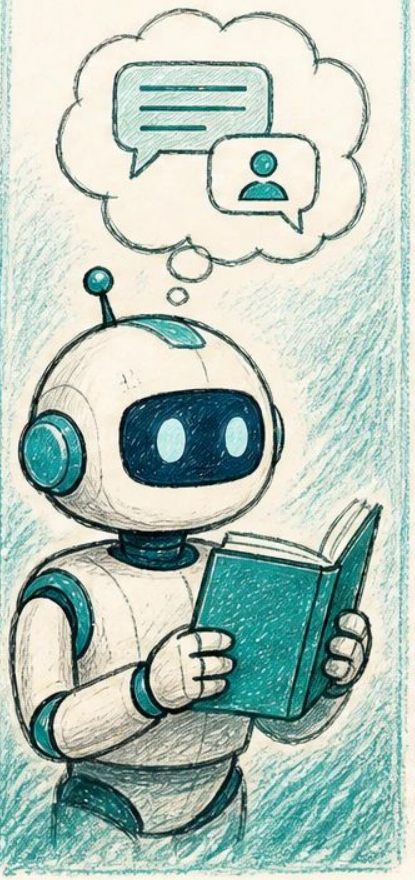
The personality the tool should adopt to accomplish tasks. Custom instructions that personalize the outputs to each task.



4

MEMORY

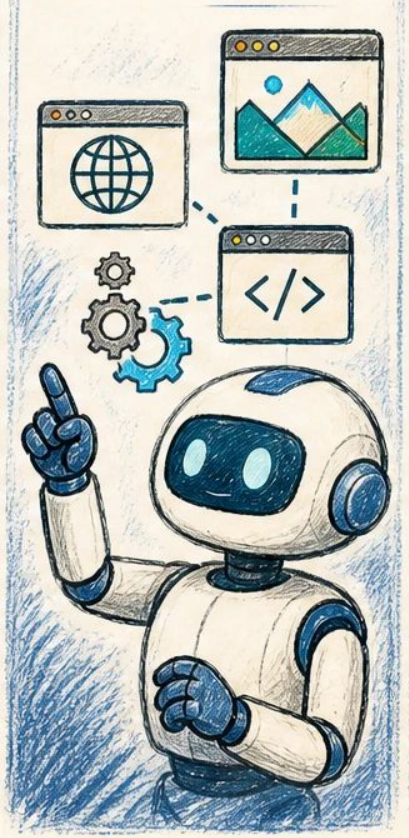
What the AI remembers about you and your past conversations, adds it as context to conversations.



5

TOOLS

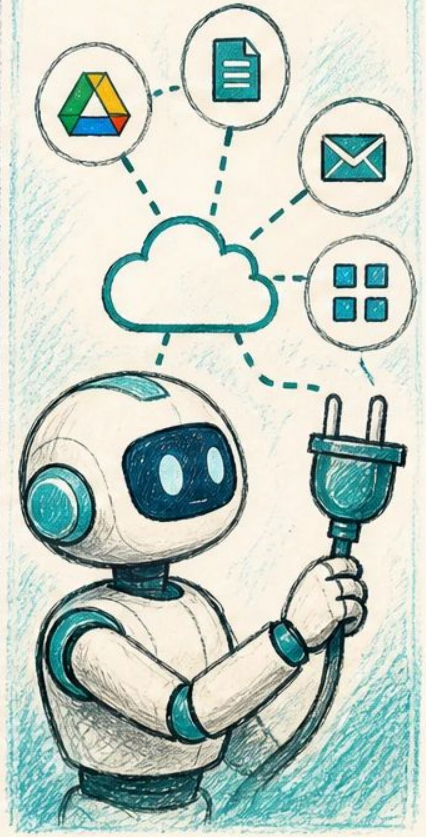
The model can use different applications like a browser, image generators or write its own code to take care of more complex tasks.



6

CONNECTORS

How it reaches external services: Google Drive, Work IQ, Apps using the Model Context Protocol (MCP) Standard.



Service	Description	Provider	Available To	Risk Category	Use Cases	Cost
Stanford AI Playground	Simplest entry point. Basic chat across many models (OpenAI, Google, Anthropic, Deepseek). Approved for low & moderate-risk data.	Stanford	Staff, Faculty, Students, Postdocs	Low Risk Moderate Risk High Risk (No PHI)	Multi-Model Chat	Free
Claude for Education	Best at getting work done with top-tier UI/UX coding. Strongest harness for Pro users. Supports skills, plugins, and MCPs.	Anthropic	Staff, Faculty, Students, Postdocs	Low Risk Moderate Risk High Risk (No PHI)	Chat and Collaboration	Free, Paid
Google Gemini Enterprise	Deep Google Workspace integration. Nano Banana for image work, scheduled actions, and strong multimodal capabilities.	Google	Staff, Faculty, Students	Low Risk Moderate Risk High Risk (With PHI)	Chat and Collaboration	Free, Paid
OpenAI ChatGPT Edu	Most popular by user count. Excellent memory, great writing, superb model, strong coding, and agentic OS.	OpenAI	Staff, Faculty, Students	Low Risk Moderate Risk High Risk (With PHI)	Chat and Collaboration	Free, Paid
M365 Copilot	Connects to Microsoft Work IQ. Works seamlessly with Word, Excel, PowerPoint. Enterprise-grade security and compliance.	Microsoft	Staff, Faculty	Low Risk Moderate Risk High Risk (With PHI)	Office Integration	Paid

Platforms with Massive Upgrades

Stanford's AI pilot gives faculty, students, postdocs, and staff campus-wide access to three major AI platforms.

- **Three platforms available.** ChatGPT Edu, Google Gemini Enterprise, and Anthropic Claude for Education.
- **Simple access process.** ChatGPT and Claude require a request form and single sign-on; Gemini activates automatically with a Stanford Google account.
- **Data protections built in.** Conversations aren't used to train models, and most tools are approved for low-, moderate-, and high-risk data.
- **PHI has limits.** ChatGPT and Gemini handle protected health information, while Claude requires the AI API Gateway for PHI use.
- **Users stay responsible.** Everyone must verify AI outputs and follow Stanford's privacy, data protection, and academic integrity policies.

PICK THE RIGHT MODE FOR THE WORK

CHAT / WORK

Your everyday productivity partner

- File aware
  reads folders
- Runs a sandbox
- Controls
  native apps
- Uses Skills
  and Plugins



CHAT / WORK | CODE

CODE

Built for developers



- Engineering work with full repo awareness
- Available as a CLI
- In the App
- Or in the Cloud



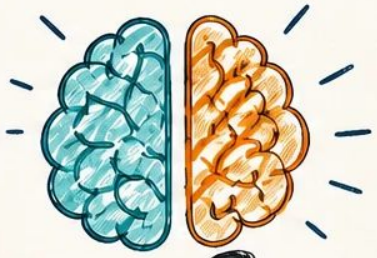


SAME FOUNDATION

Memory | Skills | Projects



CHOOSE YOUR MODEL AND YOUR EFFORT



ONE CHOICE. TWO LEVERS.

Model

Choose a model

YOU CHOOSE the model

YOU SET the effort level

FAST

Great for everyday tasks and simple questions.



REGULAR

Good for balanced answers and everyday work.



DEEP

Best for complex problems and multi-step work.



PICK YOUR EFFORT LEVEL

More effort = deeper thinking. You choose.

LOW — HIGH

HOW IT WORKS

LOW EFFORT →

HIGH EFFORT →

YOU CHOOSE THE MODEL. YOU SET THE EFFORT. BETTER RESULTS.



Right model.
Right effort.
Better answers.
Every time.

TWO SPEEDS OF TRUST

Cowork lets you decide how much rope you get between steps.



1 ASK BEFORE ACTING

Default. You lay out the plan. You approve. Then it runs.



TRADE OFF:
Approval prompts catch problems early but slow you down.

VS.

2 ACT WITHOUT ASKING

Faster. You execute the whole plan without checking in.



TRADE OFF:
No prompts feel like flow but raise the risk of prompt injection from a bad file or page.



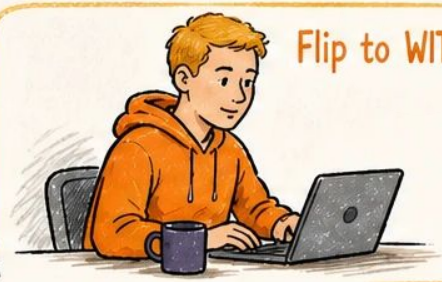
HOW TO CHOOSE

Default to ASK BEFORE ACTING for:

- New tools
- Sensitive files
- Anything unfamiliar

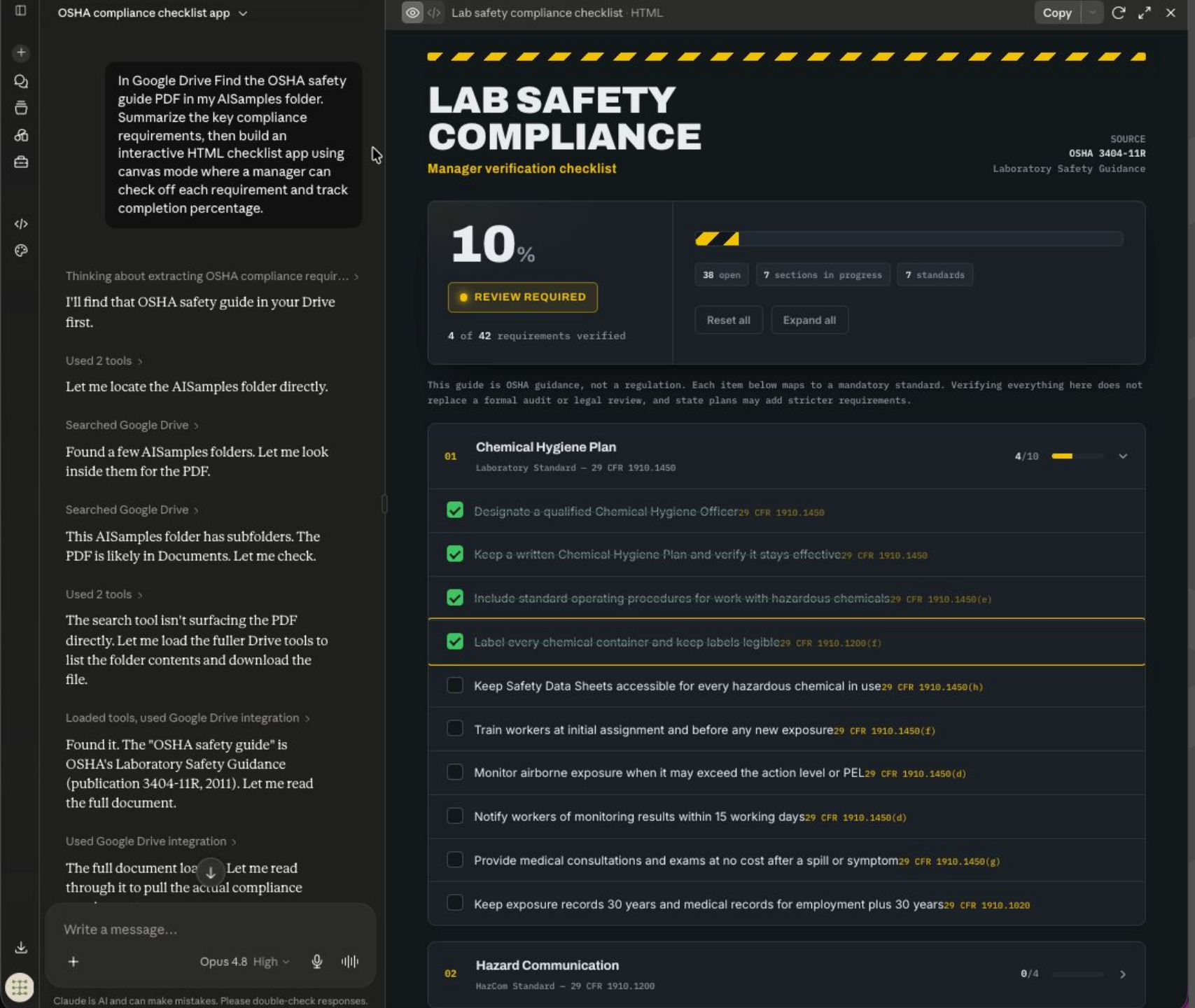


Flip to WITHOUT ASKING only when:



you are watching the screen.





Connectors & Extensions

Most AIs can pull data directly from your Google apps instead of requiring file uploads. Type in the prompt bar to connect to a specific app. (OSHA Guide)

```
In Google Drive Find the OSHA safety guide PDF in my AISamples folder. Summarize the key compliance requirements, then build an interactive HTML checklist app using canvas mode where a manager can check off each requirement and track completion percentage.
```

- Through connectors, they can find documents, extracts the content, and uses it to build something new in Canvas.
- If you use Gemini, it can use Workspace extensions Search, summarize, and pull content from Google products.
- Built-in extensions include Maps, Flights, Hotels, and YouTube Settings > Extensions).
- Claude on the other hand, has the most comprehensive types of connectors.

Podcast download analysis and trends

in the first 25 episodes to about 11,400 in the last 25.

Let me prep the data, then build the page around that finding.

Architected interactive HTML dashboard embedding podcast analytics …

Data is ready. Now I'll build the page. The design leans into the subject: a podcast analytics console styled like an audio level meter, where each episode is a bar in the signal and the rising levels across the run *are* the growth story.

Validated HTML structure and syntax integrity

Let me verify the file is well-formed before handing it over.

Ran a command, read a file

Here's the page. Three things you asked, answered straight:

Most downloaded episodes. The leader is Episode 93, "ML accelerating climate modeling projects," at 23,331 downloads. The full top ten sits in the leaderboard section, with runtimes listed so you can eyeball the lack of pattern. They range from 28 to 33 minutes, which is the same range as the duds.

Duration vs downloads correlation. There isn't one. Pearson r is -0.042 with a p-value of 0.68, which is statistically indistinguishable from zero. The scatter plot proves it visually: the cloud of dots doesn't tilt, and the best-fit line is flat (a hair downward, if you squint). Length is noise.

The thing the data actually screams. Downloads track *episode order* at r = 0.887. The show is compounding at roughly 6% per episode. First 25 episodes averaged 165 downloads, last 25 averaged 11,368. That is the growth chart on the page, styled as a rising level meter since that felt right for a podcast. Flip it to log scale to see the early episodes that get flattened on the linear view.

Two interactive bits worth poking: the **Color by era** toggle on the scatter makes the real story pop (every high dot is just a recent episode, sprayed across all runtimes), and the **Log scale** toggle

Podcast analytics · HTML

Copy

Download as HTML

Publish artifact

SIGNAL

PODCAST ANALYTICS // 100 EPISODE

DOWNLOAD ANALYSIS

Length never moved the needle. Time did.

Across 100 episodes the show pulled 457,641 total downloads. The first 25 episodes averaged 165 downloads. The last 25 averaged 11,368, roughly 69 times higher. Runtime stayed pinned between 27.9 and 33.2 minutes the entire time and explained none of it.

457,641 Total downloads	+5.9% Avg download growth per episode
-0.042 Duration vs downloads (r)	72% Average completion rate

01 The signal rising

Every bar is one episode, left to right in airing order. Color tracks intensity from quiet teal to hot orange. The climb is the whole story: downloads compound as the back catalog and audience grow.

Linear | Log scale | episode vs downloads r = 0.887

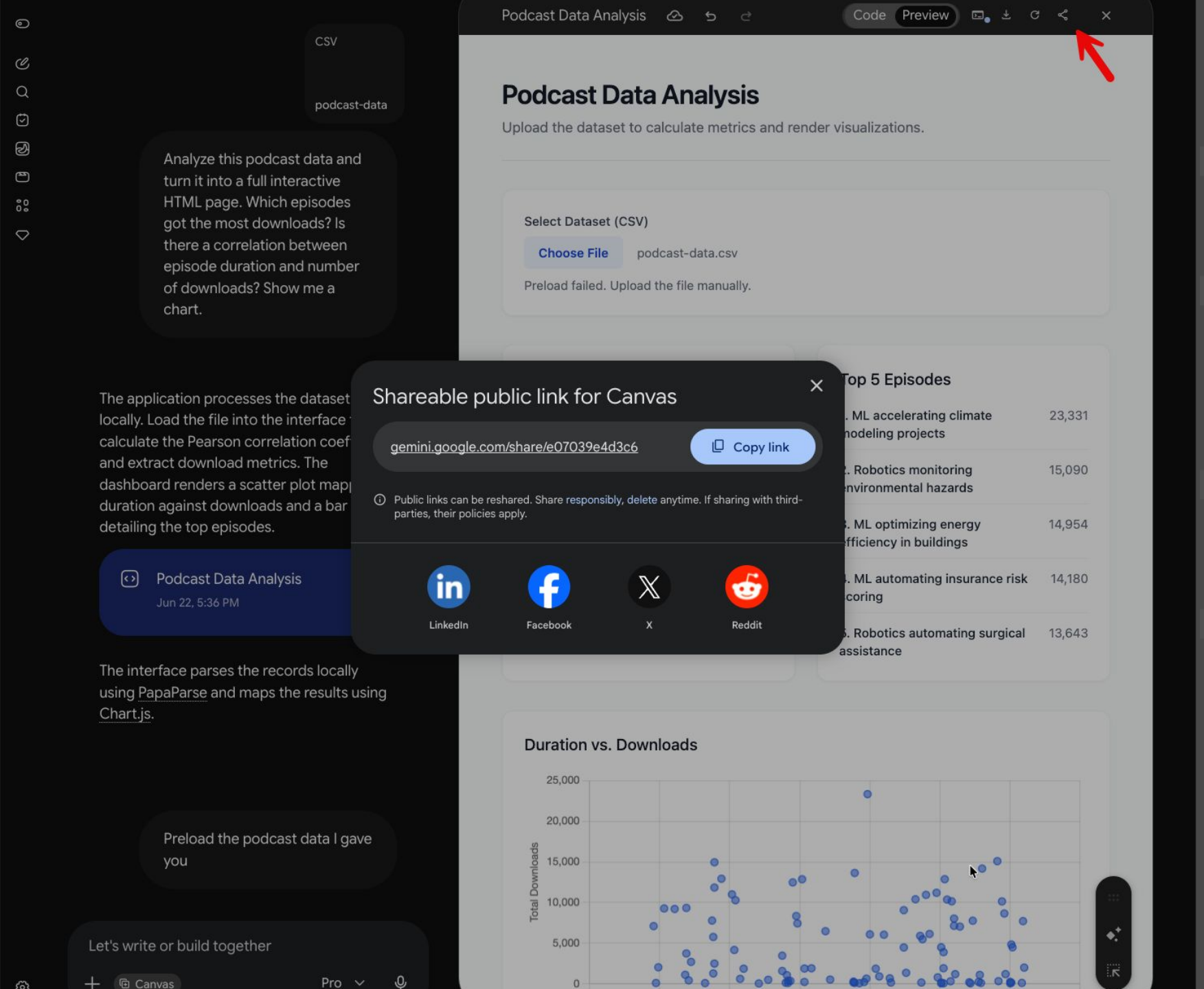
Building Dashboards Canvas/Artifacts

Chatbots process data into a shareable website. Upload podcast-data.csv to Gemini or Claude.

```
Analyze this podcast data and turn it into a full interactive HTML page. Which episodes got the most downloads? Is there a correlation between episode duration and number of downloads? Show me a chart.
```

If you're using Gemini, make sure you turn on Canvas.

- Claude will always assume that you want to build a website with the data
- Gemini and ChatGPT may assume you want to use Python. (Make sure you turn on canvas and ask for an HTML page as I did above).
- Make sure you are using the best Model and your most capable settings for best results.
- Gemini | Claude



Publishing Canvas/Artifacts

In order for others to see your work, you need to Share or Publish your artifact.

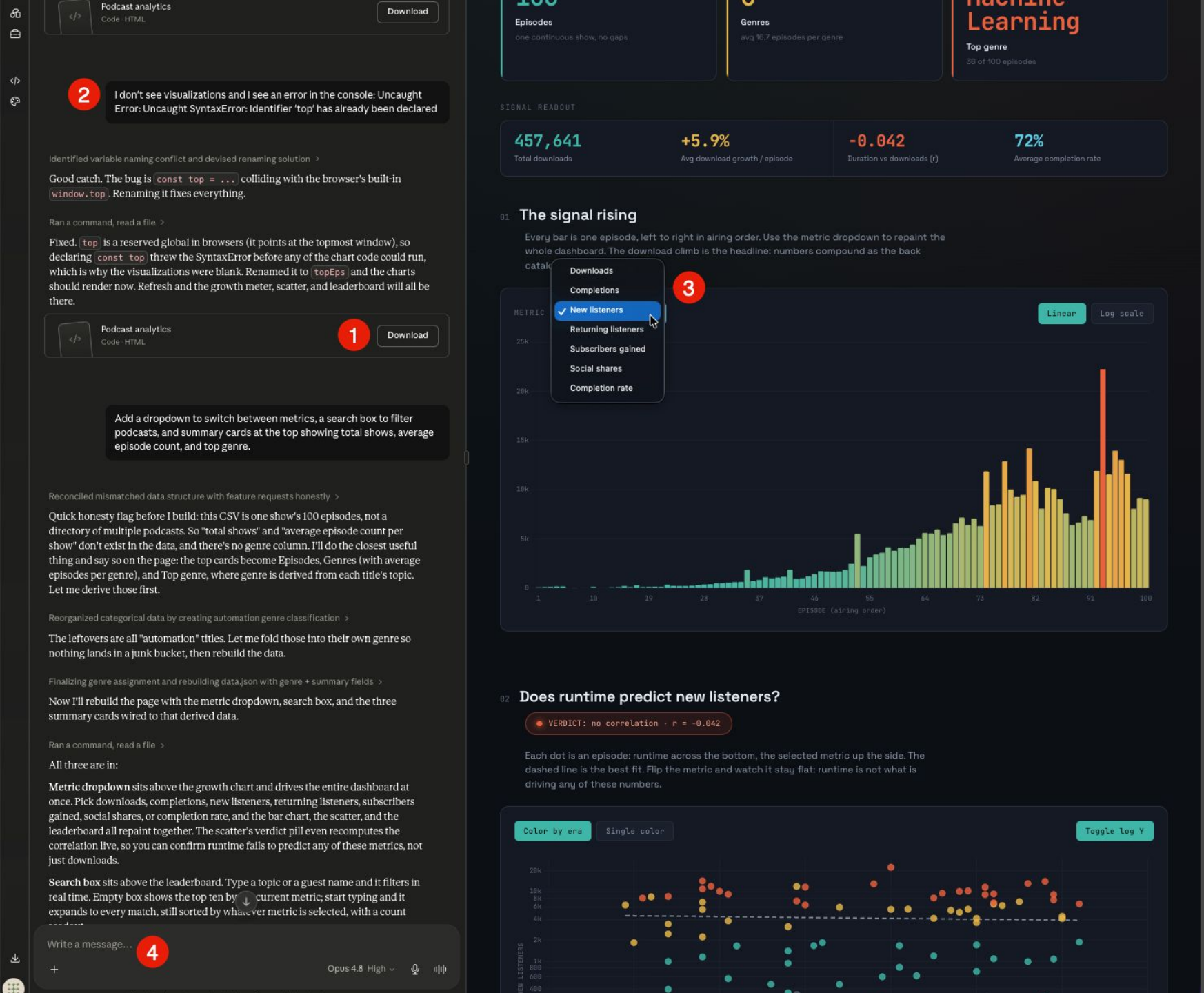
- In Claude the page is public by default.
- In Gemini, it depends on which account you're using.
    a. If you're using a corporate account with Gemini, you can use Google Drive to share with those within your domain.
- For assignments, please use a public version of Gemini or Claude, do not use a Stanford or other company's Gemini.

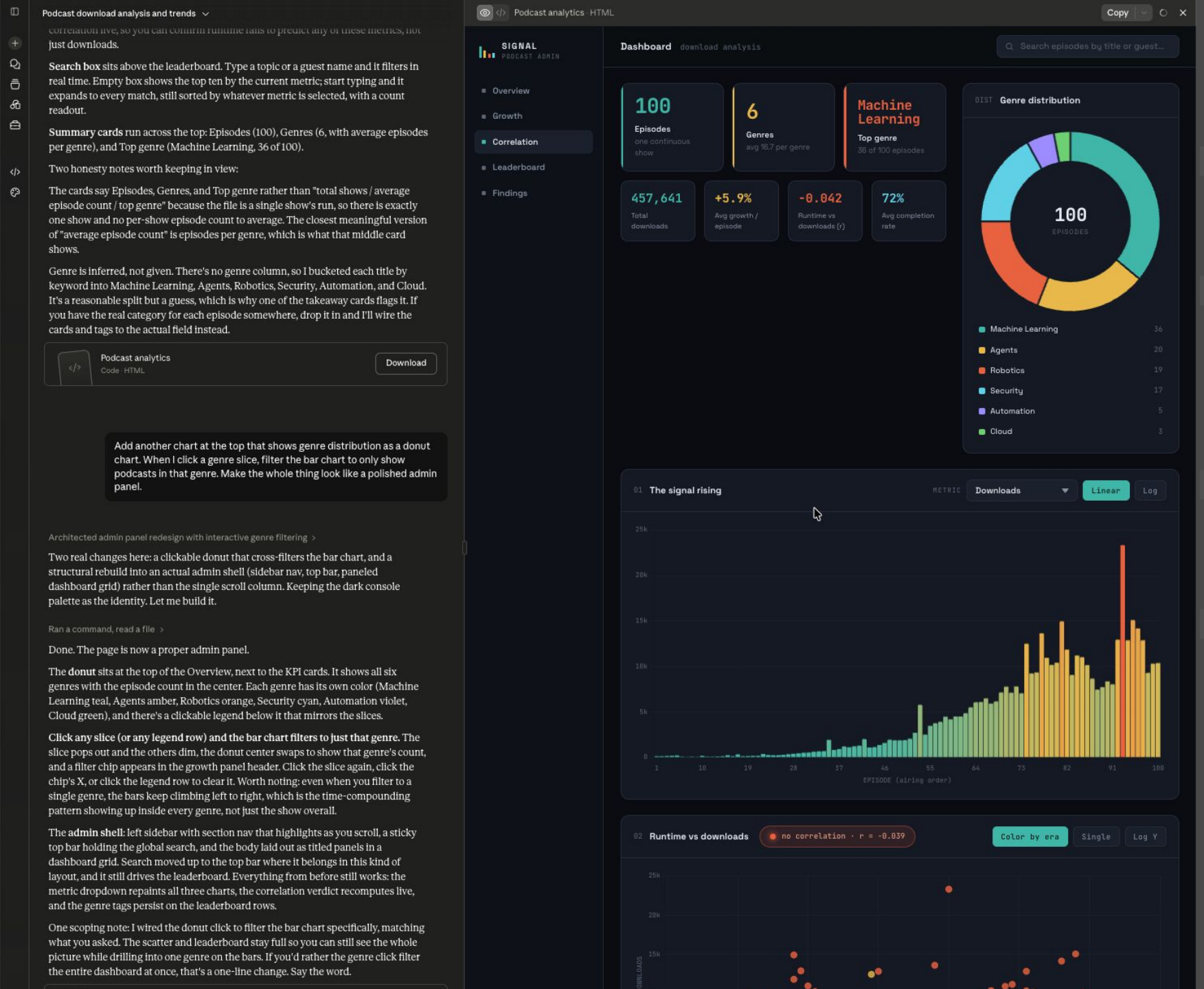
Enhancing Dashboards

Follow-up prompts refine the same Canvas output without starting over. Each iteration builds on the last.

```
Add a dropdown to switch between metrics, a search box to filter podcasts, and summary cards at the top showing total shows, average episode count, and top genre.
```

1. Every follow-up modifies the existing Canvas code, not a new file.
2. Sometimes, you will get errors, ask it to fix them for you and include screenshots or console messages.
3. Think about what controls your audience needs: dropdowns, search, date ranges, toggles (see my Vibe Glossary)
4. Be specific about what you want: name the charts, metrics, and layout. Don't be afraid to ask for suggestions.





Cross-Chart Interactivity

Connecting charts to each other turns a collection of visuals into a better tool.

```
Add another chart at the top that shows genre distribution as a donut chart. When I click a genre slice, filter the bar chart to only show podcasts in that genre. Make the whole thing look like a polished admin panel.
```

- Linked charts let users explore data by clicking, not just reading
- The AI handles the JavaScript event wiring between components
- Other chart types to try: treemaps for hierarchies, scatter plots for correlations, heatmaps for time patterns, radar charts for multi-variable comparisons
- You can also ask for hover tooltips, animated transitions, and click-to-drill-down on any chart



Beyond Web Pages

Canvas is not just a code tool. It sits on top of Google's full AI product suite, which means you can generate more than websites.

- Canvas output types include web pages, infographics, quizzes, flashcards, and audio overviews
- You can convert between types at any time: "turn this into a quiz" or "make an audio overview"
- Because Gemini connects to Google Maps, YouTube, Translate, image generation, and text-to-speech, your Canvas apps can tap into capabilities that would normally require separate APIs and developer setup
- Think beyond the dashboard: travel guides with embedded maps, multilingual flashcard sets, audio explainers, interactive timelines, data-driven slide decks

Ran a command, read a file >

Here's the infographic. Clean vertical poster, ranked top to bottom, one card per genre with a custom line icon (neural nodes for Machine Learning, a bot head for Agents, an arm for Robotics, a shield for Security, a gear for Automation), the genre color carried through the icon tile, count, and bar.

The five, by episode volume:

Machine Learning leads at 36, then Agents 20, Robotics 19, Security 17, and Automation 5. The proportional bars make the gap obvious: Machine Learning is nearly double anything else, and Automation is a thin sliver. Cloud (3 episodes) misses the top five.

I added a second number to each card worth a glance: average downloads per episode. It tells a different story than volume. Machine Learning is the most *produced* genre but the lowest performing of the big four at about 3,973 downloads per episode, while Robotics, the show makes fewer of them, averages the highest at about 5,952. So volume and pull are not the same thing here.

The honest caveats are baked into the footer: genre is inferred from titles, and since this is a single show, "count" is the real measure rather than an average across shows. If you want a second version that ranks by average downloads instead of volume, or one sized for a specific format (LinkedIn 1080 square, story, slide), tell me the dimensions and I'll set it.

Genre infographic
Code · HTML
Download

Write a message...

Opus 4.8 High



Infographics

Canvas can turn your data into a visual summary designed to be shared.

```
Create an infographic comparing the top 5 podcast genres by average episode count. Use icons, bold colors, and a clean vertical layout.
```

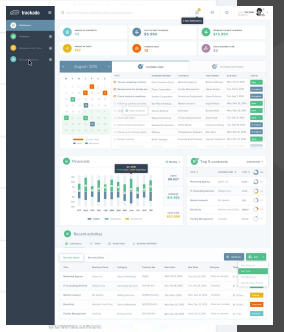
- Infographics are static visual layouts, not interactive apps. Good for reports, social media, and presentations.
- You can ask for specific styles: "corporate", "hand-drawn", "minimalist", "magazine layout"
- Unlike dashboards, infographics don't need JavaScript. The AI generates HTML/CSS only, which makes the output more predictable.
- You can also ask for it to make it into an svg or a PNG.

Steering

Upload screenshots, mockups, or color swatches to influence how the AI designs.

```
Here's a screenshot of a dashboard I like. Rebuild my podcast dashboard to match this visual style, including the card layout, font choices, and color palette.
```

- Rewind to a previous render to start from that spot
- Say "match this look" , visual style, etc.
- Paste an image or list hex codes (e.g., "#1a1a2e, #16213e, #0f3460, #e94560").
- Ask to "Use the Spotify design language".
- [Designmd.ai]
- [99designs: Dashboard Inspiration]
- [Dribbble: Dashboard UI]
- [Behance: Dashboard Projects]
- [Muzli: Dashboard Examples for 2026]



88% drops.

The per-genre gain-and-loss story lives in the hover. Each link tells you its share of the stage and, more usefully, the conversion: completions per new listener for the middle band, and percent of completions shared for the right band. That is where the genres separate. Robotics and the two small genres, Cloud and Automation, punch above their weight at sharing (around 13 to 15% of completions shared), while Security lags at about 10.5%. Machine Learning is the giant by volume but middling on share conversion at 11.75%. So the funnel is not uniform: some genres convert finishers into sharers far better than others, and that is the real signal under the raw totals.

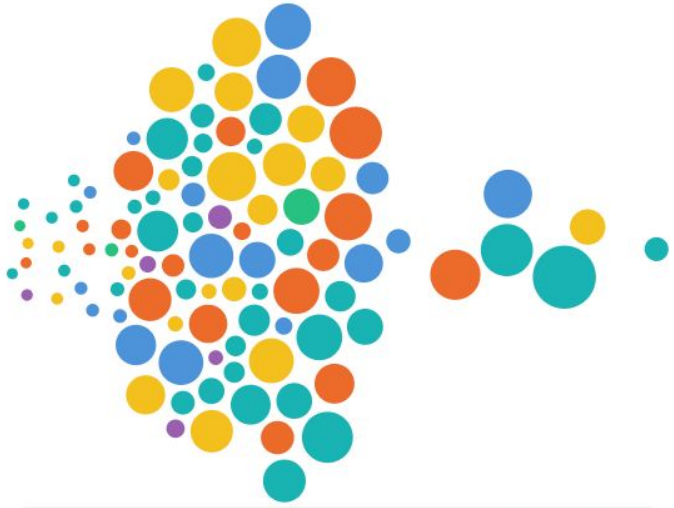
It also ties into the rest of the dashboard. Click a genre in the donut or its legend and the Sankey fades every other genre, same as the network graph, so you can isolate one genre's path through the funnel. New "Funnel" item is in the sidebar nav.

One honest caveat worth stating plainly: this is a genre-to-stage branching view, the structure you picked, so the three stages are parallel destinations rather than a strict chained flow. The funnel drop-off is legible through the relative node sizes and the conversion tooltips rather than through links physically narrowing from one stage into the next. If you ever want the true chained version, New to Completions to Shares as three sequential columns with each genre as a colored ribbon threading through, that is a different build and I can do it as a second diagram.

Write a message

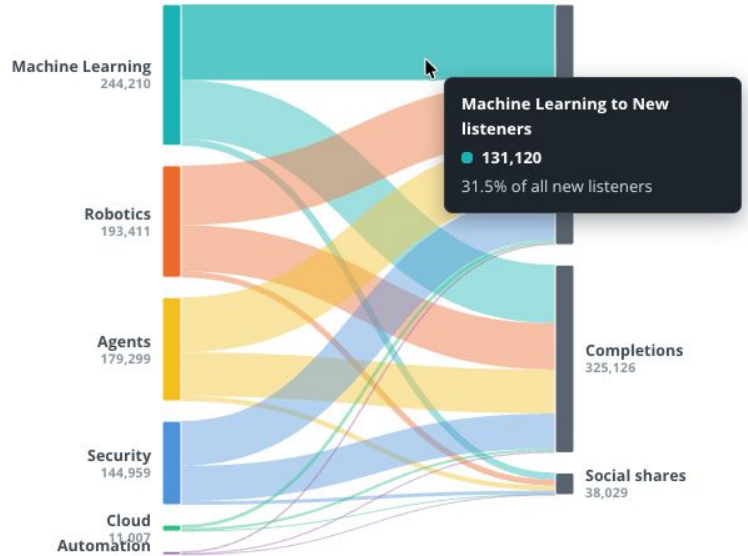
Overview
Growth
Network
Funnel
Leaderboard
Findings

Re-cluster



Engagement funnel by genre

band width = raw totals · new listeners to completions to shares



Libraries

The default Chart.js works fine, but you can ask for any front-end library by name. D3.js unlocks chart types that Chart.js can't do.

```
Replace a chart with a D3 force-directed graph. Each podcast is a node sized by downloads. Group them by genre with color coding. Podcasts with similar new-to-returning listener ratios should cluster closer together.
```

```
Add a Sankey diagram that shows how listener engagement flows from new listeners to completions to social shares, broken down by genre. Show where each genre gains or loses audience through the funnel.
```

- **Chart.js**: Simple, fast, and good defaults. Best for bar, line, and pie charts.
- **D3.js**: Complex visualizations. Browse the D3 Graph Gallery for ideas.
- **Plotly**: Interactive scientific charts with features like zoom, hover tooltips, and 3D.
- **shadcn/ui**: A polished component library for interfaces and form controls.

File Uploads

A CSV drag-and-drop turns your dashboard from a one-time demo into a reusable tool.

```
Add a file upload area at the top of the dashboard. When someone drops a CSV file onto it, parse the data and update all the charts automatically. DO NOT require file uploads, start by showing the default data first.
```

- The upload runs entirely in the browser using the FileReader API. No server needed.
- The dashboard becomes a template anyone can reuse with their own data.
- This is the line between "a thing I made once" and "a tool I can share."

Podcast download analysis and t...

Podcast analytics · HTML

Copy

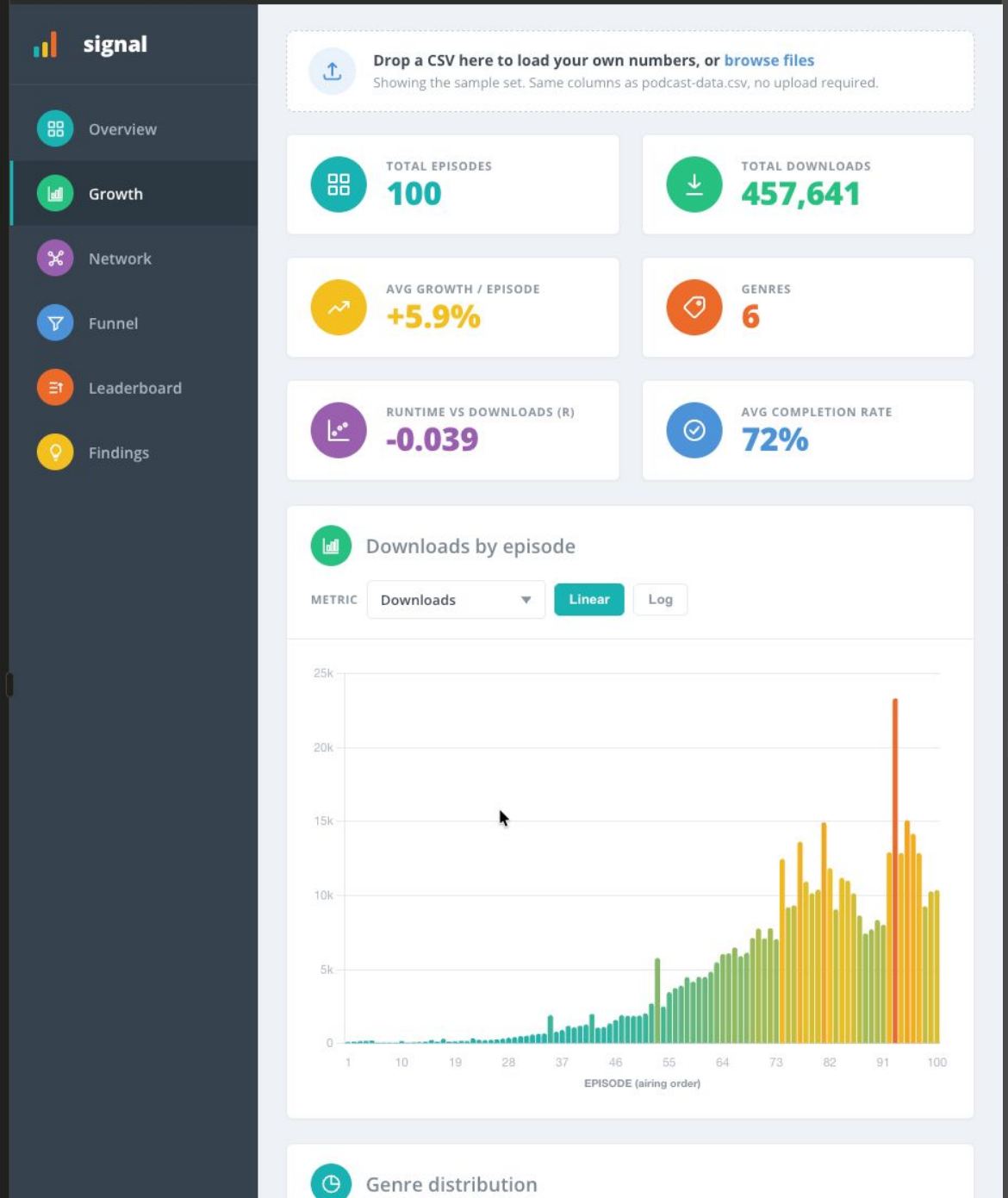
loads first with zero interaction required, so nothing about this gates the dashboard behind an upload.

Drop a CSV onto it (or click to browse) and the whole thing recomputes live: KPIs, the growth bars, the genre donut, the force network, the Sankey funnel, the leaderboard, and the findings cards all redraw from your numbers. The status line confirms the load ("Loaded yourfile.csv. N rows in, every chart redrawn"), and a "Back to sample" button appears so you can flip back to the demo set.

The part that took the real work: every number the dashboard shows was precomputed in Python and baked in. To make uploads honest, I ported that math to run in the browser. So on upload it parses the CSV (using d3's RFC-compliant parser, which handles the quoted description fields with commas), re-derives genres from titles, recomputes the regressions, the growth rate, the genre splits, and even the correlation p-value via a proper Student's t calculation rather than faking it. The numbers you get from an upload are computed the same way the sample's were.

It expects the podcast-data.csv shape: episode, title, downloads, completion_numbers, new_listeners, returning_listeners, social_media_shares, and duration are required; guest and subscribers_gained are optional. Duration parses H:MM:SS, MM:SS, or plain minutes. If a file is missing required columns it tells you exactly which ones and keeps the data you were looking at, so a bad drop never blanks the screen.

Two honest limits worth knowing. Genre is still keyword-guessed from titles, so a CSV from a different domain will mostly land in the default bucket until you give it a real genre column (say the word and I'll wire one

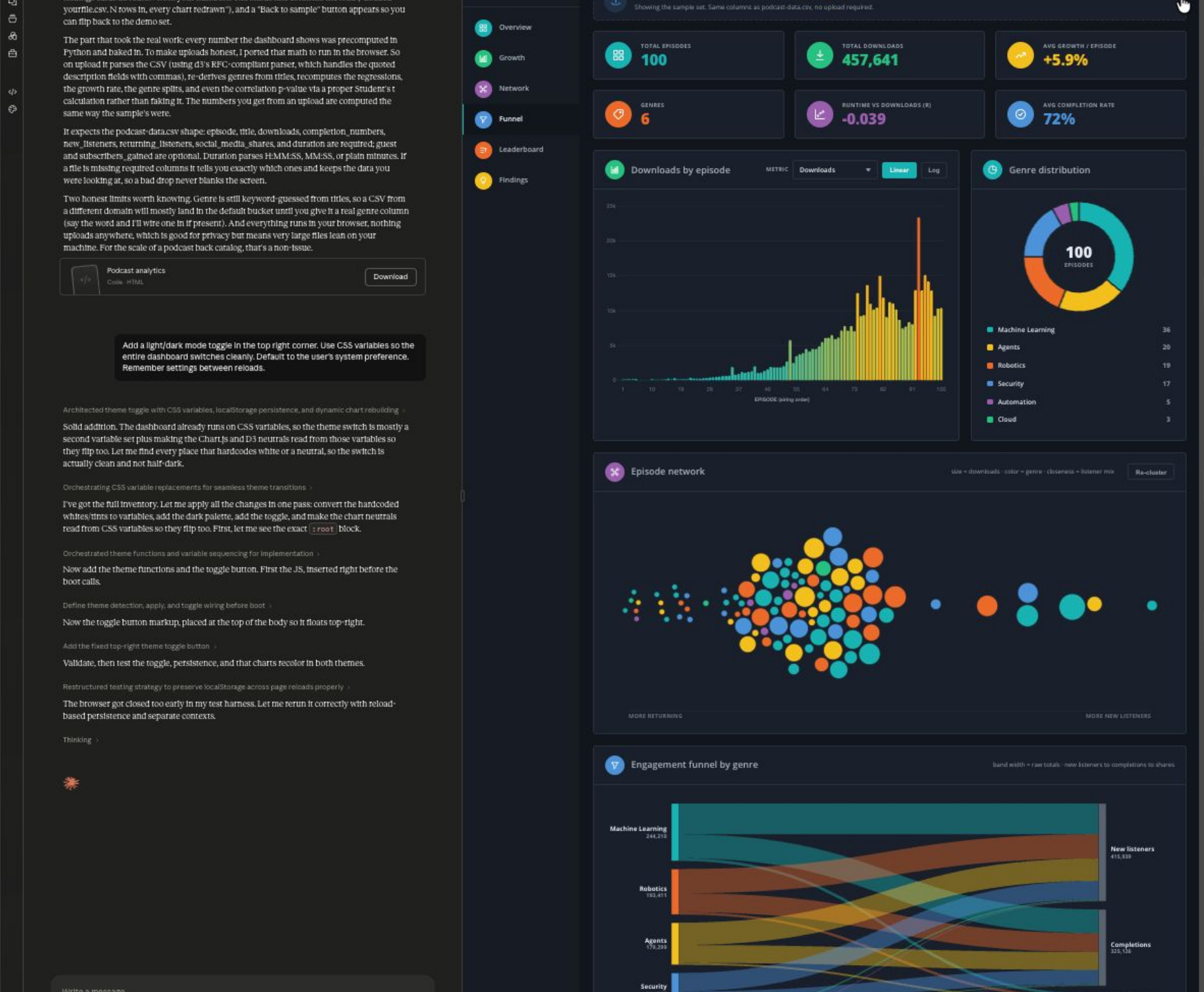


Modes & Local Storage

Asking for theme controls teaches the AI to restructure CSS, not just swap colors.

```
Add a light/dark mode toggle in the top right corner. Use CSS variables so the entire dashboard switches cleanly. Default to the user's system preference. Remember settings between reloads.
```

- CSS variables let one toggle change every color across the whole page
- "Default to system preference" uses the `prefers-color-scheme` media query
- Remember between reloads: local storage stores settings to remember the user. That can be applied to other settings.



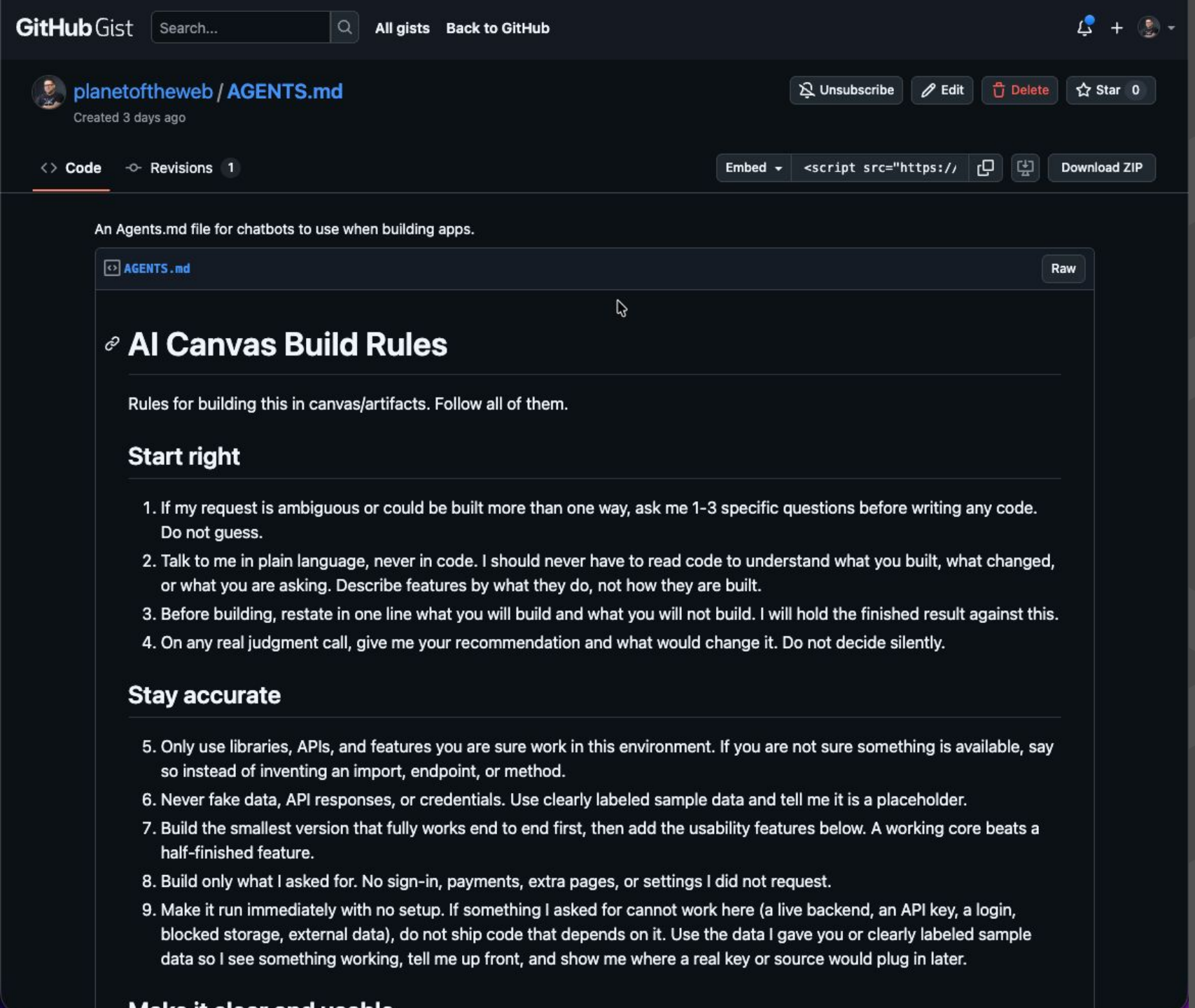


AI Features

You can embed AI-powered features directly inside a Canvas app using the Gemini API.

```
Add an AI assistant panel to the sidebar. When I type a question about the data, it should analyze the current dataset and respond with insights.
```

- Canvas apps can call the Gemini model to add chat, summarization, or analysis features
- It can sometimes also call upon other AIs' for things like voice input or image generation. Try it!
- The AI writes the integration code for you, making it an easy add.
- Some of these might be rate limited and subject to availability, they may require a certain level of account.



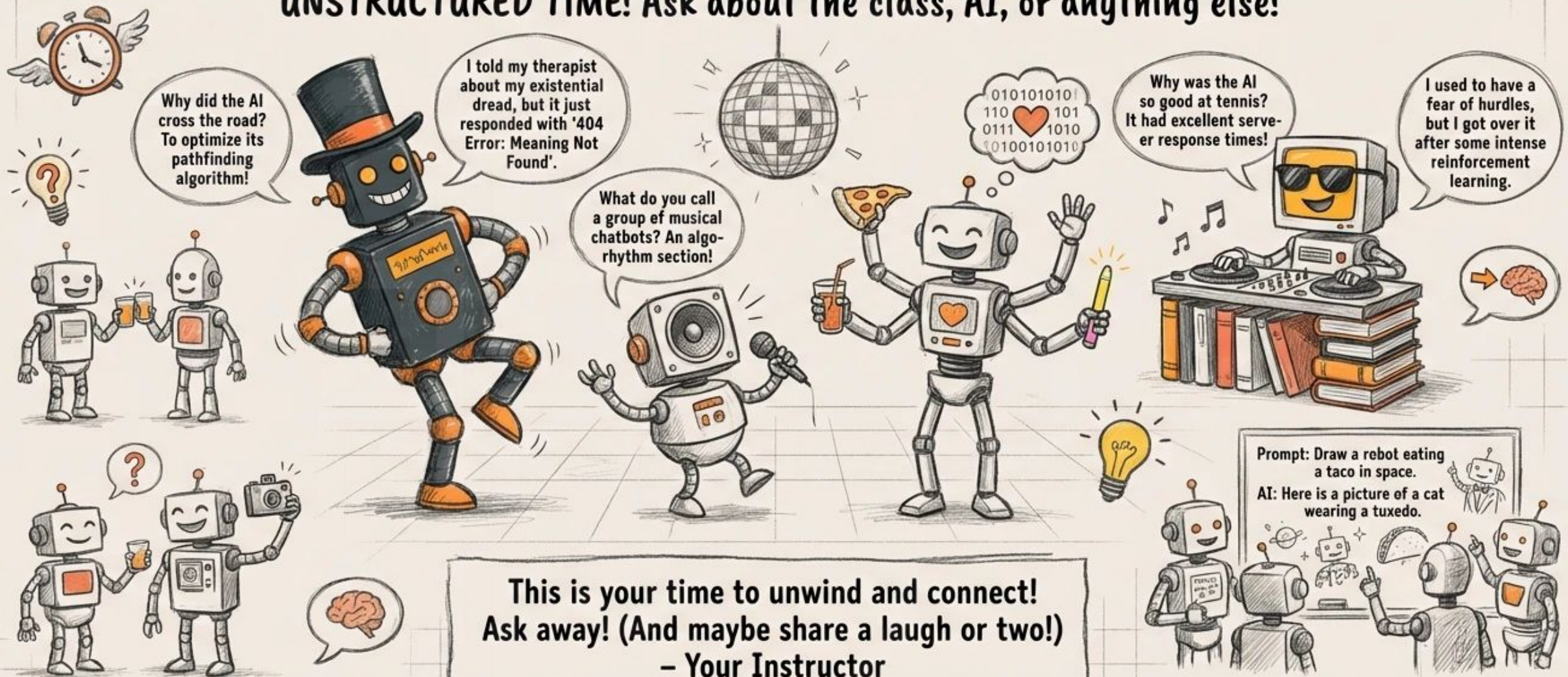
Guiding Files

You can use certain files or prompts to help Claude or other AIs stay focused and prevent drift. You can copy or download these here.

- **Start right:** Ask clarifying questions when ambiguous, restate scope before building, communicate in plain language, and surface judgment calls with recommendations.
- **Stay accurate:** Only use libraries and APIs known to work here, never fake data or credentials, and build the smallest fully working end-to-end version first.
- **Make it clear and usable:** Every screen explains itself with titles, summaries, and labels, plus tooltips on jargon so first-time viewers understand without asking.
- **Design for everyone:** Support light/dark mode, full responsiveness from phone to desktop, and keyboard accessibility with AA contrast and reduced-motion respect.
- **Handle data well:** Render data immediately instead of empty upload screens, prefer drag-and-drop, and apply newly added data live without gating the dashboard.
- **Don't drift:** Change only what's asked, never quietly remove working features or rename untouched code, and keep one source of truth for each piece of state.

AMA (ASK ME ANYTHING) AFTERPARTY: THE CLASS CONCLUSION

UNSTRUCTURED TIME! Ask about the class, AI, or anything else!



This is your time to unwind and connect!
Ask away! (And maybe share a laugh or two!)
– Your Instructor